MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (90 minutes) December 7th, 2013
Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. (a) Suppose that X is a Cauchy random variables with density

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

Find the density of $Y = 1/X$.

(b) Suppose that W_1, W_2 are independent exponential random variables with density

Let $U = \min\{W_1, W_2\}$ and let $V = \max\{W_1, W_2\} - \min\{W_1, W\}$. Find the joint density of *(U,V)*. Are U and V independent? Justify your answer.

2. (a) Twenty five people are at a party. Each person writes their name on a piece of paper and places the paper in a blank envelope. The envelopes are then randomly distributed to the people at the party, so that each person receives one envelope. What is the expected number of people to receive the envelope that contains their own name?

(b) One evening, Bob does laundry and washes n pairs of socks (so there are $2n$ socks in total). At the end of the evening, when Bob is folding clothes, he discovers that some of his socks have been lost and only $k \leq 2n$ socks remain. Assuming that each individual sock was equally likely to be lost, find an expression for the expected number of the original pairs of socks that Bob manages to recover.

3. Let $F(x)$ be a continuous distribution function (cdf) and let U be a uniform(0,1) random variable. Show that $F(x)$ is the cdf of the random variable $X = F^{-1}(U)$.

4. Let $X_1, \ldots, X_n$ be a random sample from a Uniform(1, λ)

(i) Write down the density function of X_i and the likelihood of λ.
(ii) Give the MLE of λ and show that it is a biased estimator of λ.